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MILITARY SPECIFICATION

CONVEYOR, BELT: PORTABLE

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable, pneumatic tire, diesel-engine-driven, elevating-troughing, 50-foot, 300 tons per hour belt-type conveyor.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gage Polyolefin.
QQ-S-781	- Strapping Steel and Seals.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting.
TT-V-121	- Varnish, Spar, Water-Resisting.
UU-T-81	- Tags, Shipping and Stock.
VV-F-800	- Fuel Oil, Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3910

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PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

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MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic and Associated Equipment.)
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-T-740	- Trucks; 5 ton, 6 x 6 Military Design.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105	- Lubricating Oil, Gear, Multi-Purpose.
MIL-C-3774	- Crates, Wood; Open (12,000- and 16,000-Pound Capacity).
MIL-C-5501	- Cap and Plug, Protective, Dust and Moisture Seal.
MIL-M-7866	- Molybdenum Disulfide, Technical, Lubrication Guide.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-C-13486/12	- Cable, Special Purpose, Electrical, Low Tension, Heavy Duty Multiple Conductors Unshielded.
MIL-P-13983	- Paint, Temporary, Lusterless, Gasoline Removable.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-P-46093	- Primer Coating, Synthetic (for Brake Drums).
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.

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STANDARDS

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment.
MIL-STD-642	- Identification Marking of Combat, and Tactical Transport Vehicles.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1400	- Engines, Gasoline or Diesel, Methods of Test.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Applications.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS 35000	- Battery, Storage, Lead-Acid, Waterproof.
MS 75020	- Connector, Plug, Electrical - 12 Contact, Intervehicular, 24 Volt, Waterproof.
MS 75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 24-Volt, Waterproof.

HANDBOOK

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Specialty Products for Use in Ground Equipment System.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

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DEPARTMENT OF TRANSPORTATION (DOT)

49CFR 393 - DOT Federal Motor Carrier Safety Regulations (FMCSR).
49CFR 567 and 568 - Certification.
49CFR 571 - DOT Federal Motor Vehicle Safety Standards (FMVSS).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, 710 North Capital Street, NW, Washington, DC 20402.)

DEPARTMENT OF LABOR (DOL)

29CFR 1910 - Occupational Safety and Health Standards (OSHA).
29CFR 1926 - Construction Safety Standards.

(Application for copies should be addressed to the local DOL office or the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Section IX Welding and Brazing Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

CONVEYOR EQUIPMENT MANUFACTURERS ASSOCIATION (CEMA)

Standard No. 502 - Bulk Handling Conveyors, Troughing and Return Idlers.

(Application for copies should be addressed to Conveyor Equipment Manufacturers Association, 1000 Vermont Avenue, NW, Washington, DC 20005.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA Standards.

(Application for copies should be addressed to the American Gear Manufacturers Association, Suite 1000, 1901 North Fort Myer Drive, Arlington, VA 22209.)

THE MAINTENANCE COUNCIL - AMERICAN TRUCKING ASSOCIATIONS (TMC - ATA)

TMC - ATA Recommended Practices Manual.
TMC - ATA VMRS (Vehicle Maintenance Reporting System).

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(Application for copies should be addressed to the Maintenance Council - ATA, 1616 P Street, NW, Washington, DC 20036.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

IFI Standards.
OMFS Recommendations.

(Application for copies should be addressed to Industrial Fasteners Institute, 1505 East Ohio Bldg., 1717 East 9th Street, Cleveland, OH 44114.)

NATIONAL WHEEL AND RIM ASSOCIATION (NWRA)

NWRA Standards

(Application for copies should be addressed to the National Wheel and Rim Association, 4836 Victor Street, Jacksonville, FL 32207.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Year Book.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

ANTI-FRICTION BEARING MANUFACTURERS ASSOCIATION (AFBMA)

AFBMA Standards.

(Application for copies should be addressed to the Anti-Friction Bearing Manufacturers Association, 2341 Jefferson Davis Highway, Arlington, VA 22202.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC. (TTMA)

TTMA - Recommended Practices.
TTMA - Technical Bulletins.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., 1413 K Street, NW, Washington, DC 20005.)

AMERICAN SOCIETY OF AGRICULTURAL ENGINEERS (ASAE)

ASAE R276 - Emblem for Identifying Slow-Moving Vehicles.

(Application for copies should be addressed to the American Society of Agricultural Engineers, P.O. Box 229, Saint Joseph, Michigan 49085.)

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B20.1 - Safety Standard for Conveyors and Related Equipment.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 242 - High Strength Low Alloy Structual Steel.
A 536 - Ductile Iron Castings.
D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY

ANSI/AWS D1.1 - Structural Welding Code - Steel.
AWS B2.1 - Welding Procedures and Performance Qualification Standard for.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

RUBBER MANUFACTURER'S ASSOCIATION (RMA)

RMA IP-20-1977 - Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to the Rubber Manufacturer's Association, 1400 K Street, N.W., Washington, DC 20005.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The portable conveyor shall consist of a conveyor frame, a head end, a foot end, a 24-inch-wide conveyor belt, an elevating mechanism, and a diesel engine mounted above the conveyor frame, all mounted on a 2-wheel

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assembly for towing. The conveyor, when setting on a level surface, shall operate satisfactorily when elevated at any angle from 11 degrees through 22 degrees from the horizontal, and shall deliver 300 tons per hour (tph) of bulk material weighing 100 pounds per cubic foot when elevated at any angle in the range of 11 degrees to 18 degrees. The nominal length of the conveyor shall be not less than 57 feet between centerlines of the head and tail pulleys. The maximum width of the conveyor shall be not more than 10 feet and the overall height not more than 12 feet when the conveyor is in the travel mode. The complete conveyor, in the travel mode, shall not exceed 11,500 pounds, and the front coupler load shall be within the range of 8 to 12 percent of the total weight. The conveyor shall be capable of being towed straight ahead, moved sideways, and rotated around the foot-end ground contact point.

3.1.1 Stress limitations. The maximum stress imposed on any structural member shall not exceed 60 percent of the yield point under the following conditions:

a. Operating at rated capacity while elevated at all angles specified herein.
b. Towed over unpaved roads at a speed of 10 miles per hour (mph) while in normal travel position.

3.2 First article model. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of the conveyor belt from the approved first article inspection during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish conveyor belts conforming to this specification.

3.3 Production conveyors. Unless otherwise specified all conveyors furnished under this specification shall be physically and mechanically identical to the first article conveyor that is approved by the Government.

3.4 Safety. The conveyor shall meet the following requirements:

a. Conform to all applicable requirements of ANSI/ASME B20.1-1984 Safety Standard for Conveyors and Related Equipment.
b. Conform to all requirements of the Federal Motor Carrier Safety Regulations, except for rear bumper and as specified herein.
c. Conform to Federal Motor Vehicle Safety Standards, except as specified herein.

Safeguarding, enclosing and insulating features shall be provided for exposed components and systems which are subject to high temperatures, high pressures, electrically actuated, or inherently hazardous. Step surfaces shall provide a non-skid footing. Non-functional sharp edges, projecting points, and excessive length of fasteners shall be eliminated.

3.5 Noise limits. The noise produced by the conveyor shall conform with MIL-STD-1474 requirements when tested in accordance with specification, paragraph 4.5.3.10. The provisions of MIL-STD-1474, paragraphs 4.3 and 4.4, shall be provided only if MIL-STD-1474, paragraph 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written

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permission to exceed the 85dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, paragraph 4.3 and be readable at the distance indicated on the sign.

3.6 Oil sampling valve. An oil sampling valve shall be provided on the engine. The valve shall be manually operated and shall automatically close after release. The valve shall be made of material resistant to corrosion such that it wil not contaminate the sample. The discharge part of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way as to insure that personnel will not be exposed to danger when taking oil samples with the engine running. The location of the oil tap shall be such that when samples are taken, it shall be a true representation of the oil that is flowing while the engine is running. The valve shall be labeled crankcase.

3.7 Material. Material shall be as specified herein. All material shall be new and unused. Material not specified shall be selected by the contractor as being compatible with the intended use in accordance with the engineering standards of the earthmoving and construction equipment industry and as set forth in the publications of the American Society for Testing and Materials (ASTM), Society of Automotive Engineers including SAE Handbook (SAE), American Society of Agricultural Engineers (ASAE), American National Standards Institute (ANSI). Unless otherwise specified herein, rubber products furnished shall not exceed 8 calendar quarters in age from date of cure of the rubber to date of acceptance of the conveyor by the Government. Threaded parts shall be in accordance with the Industrial Fastener Institute (IFI) and SAE standards and recommendations.

3.7.1 Material deterioration and control. The conveyor shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

3.7.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.7.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or his/her designated representative.

3.7.4 Component ratings. All components shall be used within the manufacturer's current published ratings for this application. When applicable published ratings are not available, component manufacturer's verification or ratings must be submitted by the prime contractor to the contracting officer of the procuring activity. All components shall be certified as being compatible and properly matched to meet the specifications herein (see 4.3.1.2 and 6.2).

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3.7.5 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the conveyor may be newly fabricated from recovered materials to the maximum extent practicable, provided the conveyor produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the conveyor.

3.8 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance and repair utilizing conventional common tools and special tools associated with conveyors of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and with minimum distrubance to other components of the conveyor. Ease-of-maintenance provisions shall incorporate to the maximum practicable extent features insuring operating clearances and facilitating applicable maintenance and servicing operations. Intricate locking devices and controls with threaded fastenings which can be over-torqued by operators shall be avoided. Covers or plates which must be removed for component adjustment or for component removal shall be equipped with quick-disconnect fastenings.

3.9 Discharge height. The conveyor shall be constructed so that when on level ground, the wheels pivoted 90 degrees from the travel position, and the conveyor elevated to an angle of 18 $\pm$1 degrees, material will discharge into a receiving hopper which is 16 feet above the ground. There shall be sufficient clearance between the conveyor and the hopper so that there will be no interference between components of the conveyor and the hopper so that material will not fall short of the hopper.

3.10 Conveyor frame. The conveyor frame shall be a lattice-type truss fabricated of structural steel members securely bolted, riveted, or electric arc welded into a frame that shall meet the operating and transport conditions specified herein.

3.10.1 Belt retainers. Not less than two belt retainers shall be provided. The retainers shall be equally spaced along and shall straddle the upward run of the conveyor belt. The retainers shall be adjustable and shall be bolted to the conveyor frame.

3.11 Foot-end section. The foot-end section of the conveyor shall be provided with a hopper, takeup adjustments, a tail pulley, a dishpan-type skid, shielding to prevent entrance of material, and a detachable towing attachment.

3.11.1 Takeup adjustment. The takeup adjustment shall consist of threaded bolts to provide belt tension. The total takeup shall be not less than 15 inches including not less than 3 inches of negative takeup.

3.11.2 Tail pulley. The tail pulley shall be of the wing or self-cleaning type and shall be not less than 12 inches in diameter and not less than 26 inches long. The tail shaft shall be mounted on heavy-duty, self-alining,

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greaseable anti-friction bearings having labyrinth seals, except that taper-bore bearings are permissible when affixed to the shaft by an adapter incorporating a apered sleeve and an adjustable locking nut. The tail shaft shall extend not less than 6 inches to permit the operation of a feed. The shaft extension shall be 2-3/16 inches in diameter and shall include a 1/2-inch keyway. The keyway shall be protected by a removable safety collar. The pulley shall be crowned with a taper of 1/8 inch per foot and shall be concentric with the shaft within a total indicator reading of 0.100 inch.

3.11.2.1 Taper-type locking device (pulley and shaft). The pulley shall be provided with an inner hub which clamps to the shaft by moving axially along a taper in the outer hub. The mechanical clamping shall be accomplished by the use of a screw member or members providing for mounting and removal of the pulley.

3.11.3 Skid. A dishpan-type skid shall be mounted under the foot-end section and shall raise the section not more than 6 inches above the ground.

3.11.4 Shielding. The conveyor belt, when passing over the tail pulley, shall be protected by a sheet-metal cover attached to the conveyor frame and concentrating hopper. A barrier shall be provided to prevent material passing between the tail pulley and the conveyor belt. The sides of the foot-end section shall be covered with a removable steel sheet.

3.11.5 Towing attachment. A removable towing attachment shall be attached to the foot-end and comply with DOT FMCSR 393.70(e). The tow eye shall be in accordance with SAE J847 and shall have case hardened wear surface. The conveyor shall couple with equipment in accordance with SAE J749, at 26-inch height, and with trucks in accordance with SAE J849, through 33-inch height, when the conveyor is in the towing position.

3.11.6 Safety chains. Two safety chains fitted with safety-type chain hooks on the truck end shall be furnished. The chains shall extend a minimum of 36 inches forward of the center of the tow eye. The chains shall conform to DOT FMCSR 393.70(f) and to SAE J697.

3.11.7 Concentrating hopper. The concentrating hopper shall be mounted by bolting at the foot end of the conveyor and shall be removable. The hopper shall be three sided and skirt boarded, shall have sloping sides to prevent spillage of material, and shall be not less than 5 feet long. The space between the three sides and the conveyor belt shall be provided with adjustable and replaceable rubber flashing mounted under the hopper plates. The hopper plates shall be of steel conforming to ASTM A242 and shall be not less than 0.140 inch thick. The hopper capacity shall be not less than 7 cubic feet. The hopper shall be constructed and mounted so that when the conveyor is elevated to an angle of 18 degrees from the horizontal, the maximum height of the skirt boards shall be not more than 4 feet from the horizontal ground line and shall have sufficient clearance in the center section to receive material from discharge spouts 36 inches above the ground line.

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3.12 Head-end section. The head-end section shall consist of a drive pulley; hold-back device; driving mechanism; a belt scraper; when specified, a detachable discharge hood with fixed spout; or when specified, a swivel spout.

3.12.1 Drive pulley. The drive pulley shall be of steel not less than 26 inches in length, not less than 16 inches in diameter, crowned, lagged, and mounted on antifriction bearings. A taper-type locking device (pulley and shaft) as specified in 3.11.2.1 shall be provided. The crown shall be approximately 1/8-inch-per-foot taper to the center of the pulley. The outside surface of the pulley shall be concentric with the shaft within a total indicator reading of 0.060 inch except a total indicator reading of 0.090 inch is permissible in an area immediately adjacent to a longitudinal weld. The drive pulley shall be mounted on heavy-duty, self-aligning, greaseable anti-friction bearings having labyrinth seals, except that tapered-bore bearings are permissible when affixed to the shaft by an adapter incorporating a tapered sleeve and an adjustable locking nut.

3.12.1.2 Taper-type locking device. The taper-type locking device (pulley and shaft) shall be as specified in 3.11.2.1.

3.12.2 Hold-back device. An automatic hold-back device of the over-running-clutch type shall prevent slippage or reversal of the belt travel when the engine stops or is desengaged.

3.12.3 Discharge hood. The discharge hood shall be of the concentrating type with a fixed spout and shall be mounted on the head section. The hood and spout shall be detachable and shall handle hot mixes or dusty or fine material. Parts subject to abrasive wear caused by movement of aggregate shall be lined or faced with replaceable abrasion-resistant steel plates conforming to ASTM A242.

3.12.4 Swivel spout. When a swivel spout is specified, it shall swing through an arc of not less than 240 degrees and shall be manually controlled by a hand-wheel device mounted near the foot of the conveyor. The angle of inclination shall be adjustable. The swivel spout shall be of the open-trough type and shall be in two sections: The main section shall be approximately 5 feet long, and the second section shall be not less than 2 feet long. The side and bottom plates shall be of steel.

3.13 Conveyor belt. The conveyor belt shall be 24-inches in width. The top cover shall be heat resistant and shall be 1/8-inch thick, and the bottom cover shall be 1/16-inch thick with a skim coat between all plies. When specified the conveyor belt shall continuously handle asphaltic mixes heated to 275° F without heat damage or excessive wear. The supplier shall provide evidence that the belt conforms to the heat-resistant requirements specified herein.

3.13.1 Belt fasteners. The conveyor belt shall be provided with bolted-metal-plate, hinge-type fasteners capable of disassembly and reassembly by removing and replacing the hinge pin.

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3.14 Belt scrapers. An adjustable, replaceable, spring-loaded, single- and rubber-bladed belt scraper shall be mounted at the head-end (drive) pulley. The scraper shall cause the scraped material to fall forward and toward the main discharge. The scraper blade shall be not less than 1-inch thick, shall be split in the center and shall match the contour of the crown. A plow-type, rubber-blade scraper not less than 3/8-inch thick and 4-inches high shall be mounted inside the return run of the belt at the foot-end (tail) pulley. The scrapers shall exert uniform pressure across the full width of the belt. The scraper blades shall be neoprene rubber.

3.15 Belt housing. When specified the upper run of the conveyor belt shall be provided with a detachable sectional housing made of sheet metal not less than 0.035-inch thick and shall be galvanized with a coating of not less than 2 ounces per square foot of surface. The housing shall extend from the hopper to the discharge hood and shall protect the belt from weather. Lubrication points shall be extended through the housing where necessary.

3.16 Decking. The return run of the conveyor belt shall be protected from spilled material by a self-shedding, curved or peaked, sheet steel decking attached to the conveyor frame by bolts. The decking shall be not less than 0.035-inch thick and shall be galvanized with a coating of not less than 2 ounces per square foot of surface. The decking shall overhang not less than 3 inches on each side of the conveyor belt.

3.17 Belt carriers. The conveyor belt shall be carried on impact rolls, troughing rolls, and return rolls, conforming to the Conveyor Equipment Manufacturers Association (CEMA) Standard No. 502. The troughing rolls shall be bolted the conveyor frame. The first roll ahead of the tail pulley shall be a straight impact roll, except that the support bracket may be modified to suit the application. The first roll shall be mounted under or slightly to the rear of the foot-end concentrating hopper. The next 3 roll assemblies shall be troughing impact rolls and they shall be spaced at intervals of not more than 1-1/2 feet. The remaining troughing rolls shall be spaced at intervals of not more than 3 feet apart. The return rolls shall be spaced not more than 9 feet apart.

3.18 Elevation mechanism. The conveyor shall be provided with a mechanism which shall raise and lower the conveyor frame throughout a range from not more than 11 degrees to not less than 22 degrees of elevation in not less than 11 equal steps. The mechanism shall be hydraulically or mechanically actuated. Hydraulic rams shall be shielded to prevent damage from falling particles or other environmental conditions. A positive mechanical lock with a locking pin shall be provided to prevent the conveyor frame from lowering from any set angle.

3.18.1 Hydraulic piston rods. All hydraulic piston rods shall be chromium plated to a thickness of not less than 0.0015 inch to retard corrosion.

3.19 Drives.

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3.19.1 Chain drives. Chain drives shall be installed with idlers and shall have provision for adjustment and alignment.

3.19.2 V-belt drives. V-belt drives shall be in accordance with RMA IP-20-1977 and shall have provisions for adjustment and alignment.

3.19.3 Gear drives. Gear drives shall be in accordance with AGMA standards. Gear drives shall be totally enclosed and shall run in an oil bath. The enclosures shall conform to 3.26.4. Cast iron gears, when used, shall conform to ASTM A536. The analysis and the heat treatment shall be as recommended by the gear drive supplier. When cast iron gears are provided, the pinion shall be of steel.

3.19.4 Shafting. All shafting shall be either high-carbon steel or alloy steel. The shafting shall be turned, ground, and polished.

3.20 Chassis. The conveyor frame and elevating mechanism shall be mounted unsprung on 2, swivel-type, road-wheel assemblies equipped with 2 truck-type single tires and wheels. The chassis of the conveyor shall be suitable for transport on and off road (see 3.21.1) and on site maneuvering of the conveyor.

3.20.1 Wheel assemblies. The two road wheels shall pivot about a vertical axis and shall lock in not less than three positions: (1) a travel position with the wheels paralled to the longitudinal axis of the conveyor; (2) a radial position with the extended centerlines of the spindles meeting over the foot-end ground contact point of the conveyor; and (3) a 90-degree position with wheels in alinement and perpendicular to the longitudinal axis of the conveyor. Positive locking devices shall be provided for each pivot position. Pivoting from the travel position to the radial or the 90-degree position shall move the ground-contact surfaces of the tires toward the head end of the conveyor. Operation of the conveyor in the radial position shall result in stock piling of material in a circlar pattern about the head end, as the conveyor is periodically pivoted about the foot-end ground contact point. Operation of the conveyor in the 90-degree position shall result in stock piling of material in a straight windrow pattern as the conveyor is periodically moved sideways in a direction parallel to the stock pile. In the travel position, the road wheels shall be in correct alinement with the conveyor so that each pair of diagonal measurements from the true center of the tow eye (see 3.11.5) to the outer center of each spindle does not differ in excess of 1/8 inch.

3.20.2 Road-wheel assemblies. Each road-wheel assembly shall include the conveyor mount having a spindle, a foundation brake assembly, a hub, and a disc wheel with inflated tire. Spindles shall be of steel, corrrectly hardened and ground as recommended by the axle manufacturer, and shall be in correct alinement with the mount. Hub assemblies shall be equipped with standardized tapered roller bearings to insure replacement compatibility with the manufacturer's through-axle having a minimum 23,000-pound-capacity GAWR axles. Bearings, cups, and cones shall be all matched in accordance with the bearing and axle manufacturer's recommendations. Hubs shall be of the 10 stud type conforming to SAE J694 having a 11-1/4-inch bolt circle.

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3.20.2.1 Wheels and tires. Wheels and tires shall be in accordance with DOT FMVSS and TRA Year Book. Disc wheels shall be size 20 by 8 and shall conform to TRA No. X-4919. Radial and lateral runout shall be within industry standards for commercial truck and bus rims. The wheels shall be equipped with single, tube-type, traction lug drive wheel type tires size 11.00 by 20, minimum load range (LR) F (12 PR). The load imposed on each tire, measured at the ground in travel condition, shall not exceed the TRA Year Book recommended rated load and inflation pressure. Valve caps shall be of metal and shall be equipped with elastomer inner seals. Tire clearance in all positions specified herein shall be in accordance with SAE J683. Removable fenders and splash guards shall be furnished in accordance with SAE J682 and TTMA RP No. 41. Fenders and splash guards shall be stowed on the conveyor to prevent damage from aggregates while operating. Tire inflation pressures, as recommended by the TRA Yearbook, shall be stencilled near each tire "TP (specify) lb." in 1 inch high letters, if feasible.

3.20.3 Brake system. The conveyor shall be equipped with a weatherproof straight airbrake system, in accordance with DOT FMVSS and MCSR. The brake system shall include removable service and emergency intervehicular hoses (see 3.20.5) with gladhand couplings on each end, two flexible or rigid gladhands near the foot end, control (service) and supply (emergency) line filters, relay valve, air reservoir, air chambers, foundation brake assemblies on each wheel, and all other components and plumbing required for a complete airbrake system. All brake system components and plumbing shall be located, installed, and protected to prevent damage or malfunction from ground obstructions and debris when transported, from falling aggregate when operating, and to obtain maximum efficiency and ease of maintenance. Service and emergency gladhands shall conform to SAE J318. The gladhands shall be color coded yellow for control (service) on the curbside and red for supply (emergency) on the roadside. A metal dust cover shall be installed on each gladhand to prevent contaminants from entering the opening. The air lines shall be flexible hose type conforming to TMC-ATA RP-305 having reuseable swivel fittings. The air reservoirs shall conform to SAE J10 and shall have air storage capacity recommended for truck-trailers in SAE J813. The reservoir shall be equipped with a drain cock of the 90 degree twist type accessible to personnel from the curb side of the conveyor. Since off-road mobility is necessary, air brake chambers of the emergency-parking spring type shall not be furnished. Chock blocks in 3.20.4 shall be available for parking and automatic application of the relay valve when air reservoir pressure is low.

3.20.4 Chock blocks. Two chock blocks conforming to figure 3 installed in cradles shall be provided. The captive chain shall be hooked on the wire cradle's lower loop with the excess chain wound in a figure eight manner around the upper ears and, when uncoiled from the cradle, shall permit the chocks to be placed before or behind the conveyor tires. The chock-blocks shall withstand the conveyor being towed over the chocks without permanent deformation.

3.20.5 Intervehicular connections. The conveyor shall be furnished with a removable intervehicular electrical harness (TTMARP No. 40 plug and SAE J1067 cable) and two airbrake lines (SAE J318 glad hands on TMC-ATARP-305 hose and fittings) for connecting the truck electrical and airbrake systems to the

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conveyor systems. The airbrake control (service) line shall be located on the curbside and the supply (emergency) line shall be located on the roadside. All intervehicular connections shall be removable when the conveyor is in operation and provisions shall be made in the toolbox (see 3.31) for storage. All intervehicular connections shall be installed on the conveyor so that they extend forward a minimum of 36 inches toward the truck and extend rearward not less than 20 inches on the towing attachment, measured from the center of the tow eye. A spring clip shall be installed to hold the plug of the electrical harness to the conveyor with the harness doubled back on the towing attachment. Two dummy couplings (gladhands) shall be installed to hold the airbrake lines with the lines doubled back. Intervehicular connections shall be routed so that dropping of the tailgates of the dump and cargo trucks will not cause damage.

3.20.6 Vehicular electrical system. The conveyor shall be equipped with a commercial waterproof, 12-volt vehicular electrical system. Except as specified herein, the electrical system shall be in accordance with the following requirements as applicable for this type equipment:

a. DOT Federal Motor Vehicle Safety Standard No. 108.
b. DOT Federal Motor Carrier Safety Regulations.
c. SAE Handbook, Electrical Equipment and Lighting Section.
d. TTMA Recommended Practice RP No. 9 and RP No. 14, and RP No. 40.
e. The Maintenance Council - ATA Recommended Practices Manual.
f. Electrical component sequence and number system shall be in accord with the "Vehicle Maintenance Reporting Standards" (VMRS) of TMC - ATA as follows:

VMRS No.	Feature Requirements
VMRS 34-002	Tail and Stop Lamp: Furnish total of 4 ea. on rear of conveyor providing functions of 2 ea. outboard turn and clearance plus 2 ea. inboard stop and tail. Lamps per TTMA RP No. 14 and SAE J759, coding SAE-IST, all of the flush-mounted type either elastomer ring with hermetically sealed unit or elastomer housing with shock mount bulb socket, a snap-on or screw-on lens, and elastomer gasket. Round, 4-in. dia. size lamp installed flush in the TTMA standard 4.5-in. hole. Double filament configuration having 3 wire pigtail including a ground wire to the conveyor framing member. Install lamps in rear frame cross member, or within conveyor truss mounted boxes. Provide all lamps recessed from the conveyor edges to afford lamp protection from backing into stockpiles.

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VMRS No.	Feature Requirements (Continued)
VMRS 34-002-007	Reflectors - Safety: Furnish reflectors per TMC-ATA RP-702 and TTMA RP No. 9 plus 2 ea. additional on the front facing forward. Secure all reflectors to the vehicle with metal fasteners.
VMRS-004-021	Lamp - Marker: All running lamps (clearance not in combination lamps, identification, and side marker per SAE J592 and TMC-ATA RP-108, either flush mount type per RP-108/1 or in-line protective lens type per RP-108/3. All lamps equipped with polycarbonate plastic lenses and either hermetrically sealed shock resistant filaments or fleet service bulbs per TMC-ATA Advisory AV 1-2 having the Trade No. cited therein. All running lamps, clearance, identification, and side marker of same interchangeable make and model. Locations per TMC-ATA RP-702 and TTMA RP No. 9.
VMRS 34-003-032	Receptacle w/circuit breakers - trailers: Provide 7-way receptacle for the intervehicular connecting cable near air brake line glad hands on foot-end of conveyor in protected location. Receptacle per TMC-ATA RP-107, and TTMA RP No. 40. Install receptacle to have min. 5-degree water-drainage-slant. Either furnish an elastomer trade-tractor plug support bracket or an extended lower lip on receptacle to support the plug. Provide receptacle on a weatherproof connection box having nut and stud terminals down stream from all circuit breakers.
VMRS 34-003-036	Plug (7-conductor electrical connector) - Trailer Wiring: Provide removable front intervehicular plugs and harness assembly having two 7-way TTMA plugs and a cable harness. Assembly mate with truck per SAE J849. Plugs per TMC-ATA-107, and TTMA RP No. 40.
VMRS-34-003-037	Connecting cable-trailer-w/2 plugs (34-003-036): Provide 7-way cable per SAE J1067. Cable to reach a minimum of 36-inch forward from tow eye toward the truck.
VMRS 34-004-010	Circuit Breakers: SAE J553 circuit breakers protecting all live circuits leading from 7-way front trailer receptacle. No fuses.

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VMRS No.	Feature Requirements (Continued)
VMRS 34-004	Wiring: All wiring per SAE J163, SAE J555, TTMA RP No. 14, and the TMC-ATA RP Manual. Wire identification codes per SAE J560 or, optionally, SAE J821. Multiple circuits combined into harnesses. Wiring supported at max. 36-in. spacing with anchored elastomer ties or coated clamps to prevent whipping, chafing, and mechanical or fatigue damage. Interior of mechanical splicing devices waterproofed with RTV silicone sealant. Wire per TMC-ATA RP-110 equipped with terminals per TMC-ATA RP 112. Disconnect points for multiple circuits (in excess of 4 ea.) of the waterproof plug type or of the nut and stud junction block type. Junction blocks coated with electrical varnish after assembly. Wire-to-wire terminals secured to vehicle. On a metal plate or decal provide a wiring diagram, per TTMA Technical Bulletin No. 65.

3.20.6.1 Military vehicle electrical system. Unless otherwise specified (see 6.2), the conveyor shall be equipped with a military compatible 24-volt service system to provide towing capability for the military design trucks as well as commercial trucks. In addition to the TTMA 7-contact, 12-volt plug, the conveyor shall be equipped with the following military intervehicle, 24-volt components:

a. Military 12-Contact Plug - MS75020-1 (truck end).
b. Military Intervehicular Cable - MIL-C-13486/12.
c. Military 12-Contact Plug - MS75020-2 (conveyor end).
d. Military 12-Contact Receptacle - MS75021-1, with spring-loaded cover.

The military plug and cable shall be the same length as the TTMA plug and cable and both receptacles shall be labeled with identification plates for 24-volts and 12-volts, respectively, negative ground. The military 24-volt intervehicular cable shall be routed to the 12-contact receptacle and then into a weatherproof stud and nut type junction box, equipped with SAE J553 circuit breakers, protecting live circuits B, E, and J of the military service lamp portion. All unused leads of the cable shall be insulated and sealed watertight. The military 24-volt circuits shall be connected through replaceable solid-state electronic devices or replaceable bulbs or resistors, for live circuits, and to the conveyor's framing members, for ground circuits, in accordance with table I.

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TABLE I. Military-commercial connections.

ıction	Mil. Plug/Recpt. Terminal (circuit No.)	Connection Instructions	Comm. TTMA Wiring Indent.
Blackout Marker	A	None	-
Ser. Left Turn & Stop	B (22-461)	Connect to -	YEL (3)
Blackout Marker	C	None	-
Ground	D (90)	Connect to -	WHT (1)
Ser. Tail & Marker	E (21-489)	Connect to -	BLK(2)
		& Connect to -	BRN (6)
Blackout Stop	F	None	-
Blackout Marker	H	None	-
Ser. Right Turn & Stop	J (22-460)	Connect to -	GRN (5)
Aux. Power (Hot)	K (37)	None	BLU (7)
Ground	L (90)	Ground to Frame	-
Electric Brake	M	None	-
Electric Brake	N	none	-

With the military system installed, the TTMA marker circuits, color coded BLK (2) and BRN (6), shall be electrically isolated from each other, thus preventing 12-volt current cross-over when operated by commercial trucks through the 12-volt TTMA plug. Replaceable diodes may be used, having a minimum amperage of 150 percent of the current demand of the running lamps served. Military circuits letter coded B (461) and J (460) on the tactical trucks are passenger-car-type combination stop and turn signals, using the same lamp filament, and there is no s le stop lamp circuit. As a result, the conveyor's turn signal lamp circuits, YEL (3) and GRN (5), shall function as both turn signal lamps and stop lamps and the conveyor's normal stop circuit, RED (4), cannot be energized from the military 24-volt system input.

3.20.7 Stability. With the conveyor on level ground, the wheels in the towing position, the conveyor elevated to an angle of 18 degrees from the ground line, with no load on the belt or other ballasting, and with a vertical load of 600 pounds applied to the head end of the conveyor at the centerline of the head pulley, the down pressure of the foot end against the ground shall be not less than 100 pounds to prevent tip-over.

3.21 Transportability.

3.21.1 Performance characteristics. The conveyor shall meet the transport performance characteristics specified herein when: (1) Prepared for transport; (2) when coupled to a crawler tractor; and (3) when towed by a truck.

3.21.1.1 Highway operation. The conveyor shall travel over smooth, hard-surfaced primary highways at speeds up to 40 mph.

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3.21.1.2 Cross-country operation. The conveyor shall travel over unimproved roads and open, rolling, and hilly cross-country terrain at speeds up to 10 mph behind the truck. The conveyor shall negotiate grades up to 60 percent, mud, sand, and snow; and transport incident to intended operation sites when towed by a tractor. The conveyor shall ford hardbottom water crossings to a depth of 30 inches without prior preparation or resulting damage or malfunction.

3.21.1.3 Trailing and turning ability. The conveyor shall trail the towing vehicle without weaving. The conveyor shall follow the towing vehicle without exceeding the DOT FMCSR off-tracking requirement. Turns up to 60 degrees, as measured from straight-ahead to the right and to the left, shall not cause damage to or interference between the towing vehicle and the conveyor.

3.21.2 Braking ability. The combined service brakes of the towing and towed vehicle shall control, decelerate, and stop the combination in conformance to DOT FMVSS and FMCSR requirements. When actuated, the emergency breakaway system shall produce emergency brake application in accordance with DOT requirements.

3.21.3 Railroad. The conveyor shall withstand impact forces encountered in rail transport without damage or permanent deformation. The conveyor prepared for rail transportation, shall be within the dimensional limits shown on figure 4. The conveyor may be sectionalized providing the sectionalization and reassembly shall be accomplished in not more than 6 hours each without cutting or welding.

3.21.4 Air transport. Sectionalization for air transportation shall be within the C130 aircraft dimensional limits shown on figure 5. Sectionalization and reassembly shall be accomplished as specified in 3.21.3.

3.21.5 Slinging attachments. Permanently affixed attachments that enable the distributor to be lifted in its normal travel position shall be provided. Each attachment shall meet the requirements specified in MIL-STD-209, for type II equipment, class 1 or 3 provisions. Slinging attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.21.6

3.21.6 Equipment tiedown attachments. Permanently affixed attachments that permit fastening the distributor to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall meet the requirements specified in MIL-STD-209, for type II equipment, class 2 or 3 provisions.

3.22 Power transmission. Power transmission shall be through enclosed 90 degree, bevel-gear units connected by a drive shaft equipped with universal joints or roller chain couplings. The bevel-gear units shall run in oil and shall be equipped with antifriction bearings. The bevel-gear units shall be mounted inside the conveyor frame: One bevel-gear unit shall be near the power unit, and one shall be at the head pulley. The lower bevel-gear unit shall be driven by the power unit through a short multiple V-belt or chain, and the upper bevel-gear unit shall drive the head pulley through a short chain or V-belt

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drive enclosed in a sheetmetal housing. The drive shafts shall be mounted in antifriction, self-alining bearings conforming to standards at the Anti-Friction Bearing Manufacturers Association.

3.23 Engine. The diesel engine, including all systems, components, and accessories shall conform to MIL-STD-1410, class 1, except as specified herein.

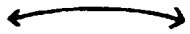
3.23.1 Power and speed rating. The power and speed rating of the engine shall be such that operation of the conveyor under any of the operational requirements and conditions specified herein will not require horsepower in excess of the net continuous horsepower rating of the engine for the applicable governed speed as published by the engine manufacturer.

3.23.1.1 Engine accessories. Engine accessories shall include the following:

a. A muffler or mufflers.
b. A 15-gallon fuel tank with drain.
c. A 24-volt electric cranking system with 18-ampere charging generator or alternator and regulator; with batteries conforming to MS35000-1, battery box, and without charging receptacle.
d. Manual cranking system.
e. Housing.
f. Instrument panel with battery-generator or alternator indicator, fuel quantity indicator, starter control, high temperature control of engine shut-off type, and mechanical hourmeter to record engine operating time at governed engine speed. All gauges and control shall be labeled in accordance with SAE J1500 (Universal Symbols) or verbage (see MIL-STD-1472, paragraph 5.5 for verbal labels).
g. The fuel fill port shall be labeled "diesel" in letters not less than 3/8 inch height.

3.23.1.2 Engine characteristics. The throttle-governor combination shall not permit speeds that will drive the belt at more than 300 feet per minute (fpm) and shall control the engine at all lower speeds for which the engine is rated for continuous operation. The governor shall be set to drive the belt at 280 to 300 fpm.

3.23.2 Clutch and clutch control. The engine shall be equipped with a hand-operated, over-center, engagement-type friction clutch and shall be capable of operation by an operator standing on the ground. The force required to engage or disengage the clutch shall not exceed 20 lb. The clutch control shall be labeled "CLUTCH



DISENGAGE ENGAGE"

The rated capacity shall be not less than 150 percent of the maximum rated engine torque.

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3.23.3 Mounting. The engine and batteries shall be mounted over the foot-end section of the conveyor frame on a base and shall have a clearance of not less than 14-inches above the belt. The engine shall be mounted so that it is level when the conveyor is elevated at an angle of 16 degrees.

3.24 Platform. A platform from which the operator may be able to crank the engine or perform routine lubrication and servicing of the engine shall be provided. The platform shall be provided with an expanded metal surface. The platform shall be hinged along one edge to enable the platform either to be raised to a horizontal position or dropped into a vertical position when not in use. Provision shall be made for holding the platform firmly in both horizontal and vertical positions.

3.25 Electromagnetic interference. The electromagnetic interference emission characteristics of the conveyor shall conform to SAE J551.

3.26 Fuel and lubrication. The conveyor shall be operable with applicable standard military diesel fuels and lubricants (VV-F-800, MIL-L-46167, MIL-L-2105, MIL-L-2104, MIL-G-10924, MIL-M-7866 and MIL-G-23827), without adverse effect on vehicle components or warranty. The lubrication intervals shall be as required by the component manufacturers. Military lubricants in accordance with MIL-STD-838 and MIL-HDBK-113 shall be used for initial fills.

3.26.1 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating as specified herein.

3.26.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534 and shall be of the threaded type. Fittings shall be located in a protected position and shall be accessible to a grease gun. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Where oil holes are applicable, they shall have spring-loaded covers. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.26.3 Pressure-release device. An automatic pressure-release fitting shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.26.4 Enclosures with integral reservoir of lubricant. Enclosures such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with manual fluid level indicators in accordance with SAE J48 and J614. Plugs shall be not less than 1/2-inch pipe size. Each enclosure shall be equipped with a magnetic drain plug. Each enclosure shall be equipped with a means for filling the enclosure with a lubricant and for draining. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure.

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3.26.5 Servicing and adjustments. The contractor shall service and adjust each conveyor for correct operation including at least the following: Adjustment of conveyor machinery, road wheel assemblies, brake system and electrical system; inflation of all tires; and complete lubrication of the conveyor with types and grades of lubricants recommended.

3.27 Fungus and moisture resistance. The electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with varnish conforming to MIL-V-173 or a varnish having 1 percent of copper-8-quinolinolate (by weight) based on the nonvolatile content of the varnish, with the following exceptions:

a. Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or functioning of the varnish.

3.28 Identification marking. The conveyor shall be identified in accordance with MIL-STD-130. All plates shall conform to MIL-P-514. Identification plates shall conform to type I, style 1, composition C, grade A, class 1. The transportation data plate and instruction, safety, warning, caution, and notice plates shall conform to type III, composition C, grade A, class 1 except as otherwise specified herein. Instruction, safety, warning, caution, and notice plates shall be furnished describing any special and important procedure to be followed in operating and servicing the conveyor and its attachments and accessories. All plates shall be securely attached to the conveyor with machine screws, bolts and nuts, or rivets and shall be furnished and mounted by the contractor.

3.29 Treatment and painting. The portions of the conveyor and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383 conforming to MIL-C-46168. The conveyor belt shall then be overcoated in accordance with MIL-C-46168 top coat of the colors specified in the camouflage patterns, with the exception that the first article may be camouflage green 383.

3.30 Vehicle marking. The marking of each conveyor shall be in accordance with MIL-STD-642. The gross vehicle weight of each conveyor and the weight of each sectionalized component shall be marked on the side in such a manner as to be discernible. Lifting and tiedown attachments shall be marked "LIFT HERE" or "TIEDOWN HERE" as applicable.

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3.31 Toolbox. The conveyor shall be equipped with a weathertight toolbox with a padlock hasp for locking. The toolbox shall be of sufficient size to secure and accommodate the special vehicle tools and on-vehicle-equipment (OVE). The toolbox shall be fabricated of metal or elastomers and shall withstand at least a 300-pound weight over any one square foot of top area. The toolbox shall be equipped with a hinged cover or a side door panel with bottom hinge and stay chains. The cover or door shall open not less than 90 degrees. The opening shall be secured by not less than two heavy-duty, tension-type, hook latches with strikers, and a hasp shall be provided for locking with a padlock having a 3/8-inch shackle.

3.32 Technical publications. Such technical publications as are specifed shall be furnished (see 6.2).

3.33 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished (see 6.2).

3.34 Government-loaned property. One M-51, 5-ton, 6 by 6 dump truck conforming to MIL-T-740 will be loaned by the Government (see 6.5).

3.35 Items to be furnished with conveyors. No shipment of conveyors shall be made unless repair parts, maintenance tools, operating tools, and accessories are included with the shipment or unless approval for shipment of the conveyors without such items has been received from the contracting officer.

3.36 Workmanship. The conveyor shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation of the conveyor. Metal used in the fabrication of the conveyor shall be free of kinks and sharp bends. Shearing and chipping shall be done neatly and accurately. Flame cutting of steel may be employed. Burned surfaces of flame-cut steel shall be ground or machined to remove all ash, burrs, and rough edges. All bends shall be made with metal dies or fixtures to insure uniformity in size and shape.

3.36.1 Welders and welding.

3.36.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by any of the following listed codes or any equivalent manufacturer's qualifications test for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/AWS D1.1 Structural Welding code - Steel
AWS B2.1 Standard for Welding Procedure and Performance Qualification or
ASME Section IX Welding and Brazing Qualifications.

3.36.1.2 Welding. Welding shall be in accordance with the AWS Welding Handbook. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without

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permanent deformation or failure when the parts connected by the welds are tested as specified herein.

3.37 Data for camouflage patterns. The contractor shall generate the following data (see 6.2).

a. Eight by ten color photos that mirror each of the views listed under 3.37.1.
b. Original illustrations of each of the views listed under 3.37.1. The illustrations shall be to 1/10 scale on E size paper. Each illustration shall include length, width, and height dimensions relative to each other and shall be clearly detailed to the extent that all discontinuities of the conveyor belt that cover one square inch or more of area, are clearly delineated to scale.

3.37.1 Camouflage pattern views.

a. Back.
b. Front.
c. Left side.
d. Right side.
e. Top.
f. Hidden views that would result in a change of a finished pattern.

Note. Hidden views are those areas requiring camouflage coloration that are not apparent from the other views. Included are variations in the appearance of the conveyor belt that result when the deployed conveyor belt is taken from a stand-y condition and placed into a ready-for-use conditon.

3.38 Government-furnished property. The following property will be furnished by the Government (see 6.5):

Item No.	Description	Identification	Quantity
1	Pattern drawings.	Camouflage patterns.	As required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Acceptability criteria. Conveyors which conform to all requirements in sections 3 and 5 of this specification and pass all applicable examinations and tests in section 4 will be considered acceptable by the Government.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specification and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The conveyor shall be examined as specified in 4.5.2. Presence of one or more defects shall be cause for rejection.

4.3.1.1 Certification. Prior to the first article examination and tests the contractor shall furnish certification as specified below that the conveyor components, and materials conform to the requirements specified herein. All certification shall be included as part of the first article test report. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Components. Certification that the following components are used within the requirements specified in 3.7.4 shall be provided: Drives (see 3.19.3), axles (see 3.20.2), Wheels and Tires (see 3.20.2.1), Power Transmission (see 3.20), Engine (see 3.23), and Clutch (see 3.23.2).

4.3.2 Tests. The conveyor shall be tested as specified in 4.5.3 through 4.5.3.8. Failure to pass any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each conveyor shall be examined as specified in 4.5.2. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each conveyor shall be tested as specified in 4.5.3.1. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Engine performance. Prior to installation of the engine in the conveyor and examination and tests of the first article model, the engine shall be tested, in accordance with MIL-STD-1400, Test Method Series 2000. Nonconformance to 3.23 and 3.23.1 shall constitute failure of this test. Upon successful completion of this test and prior to installation of the engine in the first article model the fuel system and governor shall be sealed by a

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Government representative at the settings established by this test. Seals shall also be placed by a Government representative to prevent separation of the rankcase or oil pan from the engine block and the rocker arm covers from the heads. Any changes made to these settings or removal or alteration of the seals by other than the Government representative, or without his expressed consent, will invalidate all tests previously performed.

4.5.1.1 Engine horsepower. During conduct of the 25 hour operation performance test at 22 degrees as specified in 4.5.3.2, the engine speed (rpm) and fuel rate shall be recorded at 1/2-hour intervals for the duration of this test. These readings shall be compared with the engine speed, brake horsepower and intake manifold vacuum curve, for the continuous brake horsepower condition as established during conduct of Test Method Series 2000, MIL-STD-1400 (see 4.5.1). Any fuel rate reading recorded during the 25-hour performance test (see 4.5.3.2) that is greater than the reading listed on the continuous horsepower curve established during conduct of Test Method 2300, MIL-STD-1400, at the corresponding speed (rpm), shall constitute failure of this test.

4.5.2 Examination. The conveyor shall be examined for the following defects:

101. Material not as specified.
102. Parts or components missing not identical to first article or do not function.
103. Assembly incomplete or incorrect.
104. Safety device or guards missing.
105. Workmanship not as specified.
106. Servicing and adjustments not as specified.
107. Ease of maintenance not as specified.
108. Dimensions or weight not as specified.
109. Treatment and painting not as specified.
110. Identification or other marking, operating and safety instructions, or manuals missing.
111. Lubrication devices not accessible.
112. Oil or grease seals leak.
113. Oil sampling valve not as specified.
114. Safety chains not as specified.
115. Conveyor safety not as specified in 3.4.
116. Vehicle marking not as specified in 3.30.
117. Welders certification not as specified in 3.36.1.
118. Welding not as specified in 3.36.1.2.
119. Conveyor does not elevate through the full range.
120. Conveyor not in accordance with DOT requirements.
121. Components and sub-assemblies not as specified in 3.10.1 thru 3.11.4, 3.11.6, 3.12 thru 3.12.1.2, 3.12.3 thru 3.28, 3.20.2, 3.20.4 thru 3.20.6.1, 3.21.1.3, 3.22 thru 3.24, 3.26 thru 3.27, 3.31.
122. Repair parts, maintenance tools, and technical publications not as specified.
123. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable and operating environments.

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124. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
125. Contractor does not have documentation available for identification of material, material finishes, or treatment.
126. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the conveyor.
127. Paint color not as specified.
128. Camouflage pattern data not as specified.
129. Camouflage patterns not as specified.

4.5.3.2 Performance. Operate the conveyor for 8 hours at an angle of 11 degrees, 8 hours at an angle of 18 degrees, and 8 hours at any angle in the range of 11 to 18 degrees, delivering a minimum of 300 tons per hour of the material as specified in 3.1. Operate the conveyor for 8 hours at an angle of 22 degrees with material as specified in 3.1 at maximum possible loading. Record height of discharge when angled at 18 degrees. Record on the test report the tons per hour, screen analysis, and moisture condition of the material. Feed the bulk material to the receiving hopper from a drop height of not less than 18 inches above the belt. Measure and record the maximum noise level of the conveyor and compare with OSHA requirements for an 8-hour shift. Demonstrate methods of windrowing and stock-piling. Nonconformance to 3.1, 3.2, 3.9, 3.11.7, 3.20.1 or 3.12.2, or damage to or permanent deformation of components shall constitute failure of this test.

4.5.3.3 Transportability tests.

4.5.3.3.1 Mobility tests. Tow the conveyor as specified herein for a total distance of not less than 150 miles. During all tests, the conveyor running lights shall be illuminated and turn signals operated on turns. Perform the following:

a. Before any highway and cross-country tests are conducted, measure all critical mounting points on the conveyor. After all highway and cross-country tests are completed, check and record measurement of critical mounting points and perform inspection of critical welds. Also, determine conveyor compliance with 3.1.1 stress limitations. Load, moment, and shear diagrams prepared by the contractor are an acceptable method of determination.
b. Tow the conveyor a distance of 50 miles at speeds up to 40 mph over highways.
c. Tow the conveyor a distance of 20 miles, 10 miles of trails and 10 miles of rough cross-country terrain, at safe speeds up to at least 10 mph.
d. Tow the conveyor through mud at least 12 inches deep and through creek bed type water at least 20 inches deep.
e. Tow the conveyor across highway type V-ditches not less than 18 inches deep.
f. Measure and record the deviation or off tracking of the conveyor with respect to the prime mover, when the conveyor is towed staight ahead on a level paved surface.

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g. Jack-knife the conveyor at angles up to 60 degrees from straight ahead both to the left and to the right.
h. Perform brake system test in accordance with DOT procedures, and as follows:

(1) With the conveyor stopped, perform the parking brake test by removing the chock blocks and place them, first to prevent the conveyor from rolling forward and then repeat to prevent the conveyor from rolling rearward. Do not unsnap the captive chain from the lower cradle loop. Finally, tow the conveyor over both chock blocks and inspect for damage.
(2) Test the brake system in accordance with TTMA Recommended Practice No. 12.

Failure to complete the test; inability to tow, ford, stop or function as specified herein; evidence of water in sealed or waterproof components; change in critical mounting point dimensions or evidence of any weld being defective; or nonconformance to 3.7, 3.10, 3.11.5, 3.21, or 3.21.1 through 3.21.2 shall consitute failure of this test.

4.5.3.4 Electromagnetic compatability. The preproduction model shall be tested to determine conformance to 3.25. The contractor shall furnish the contracting officer the report of test. Disapproval of the report shall constitute failure of this test. Upon approval of the reports by the contracting officer and provided all other requirements of the specification are met, the preproduction model shall be used as a model for all production conveyors. Disapproval of the report shall constitute failure of this test.

4.5.3.5 Stability. Place the conveyor on level ground with the wheels in the towing position. Elevate the conveyor to an angle of 18 degrees from the ground line and place a weighting scale under the foot end. With no load on the belt or other ballasting, apply a vertical load of 600 pounds to the elevated head-end of the conveyor at the centerline of the head-end pulley. Observe for any tendency of the foot-end to leave contact with the ground and measure the down pressure of the foot-end. Nonconformance to 3.20.7 shall consitute failure of this test.

4.5.3.6 Railroad and air-transport. Prepare the conveyor for railroad and for air transport, including sectionalizing as required and reassembling the conveyor. Nonconformance to 3.21.3 and 3.21.4 shall constitute failure of these tests.

4.5.3.7 Lifting and tiedown attachments. Lifting and tiedown attachments shall be statically tested for compliance with MIL-STD-209 strength requirements.

4.5.3.8 Tiedown attachments (dynamic). When specified (see 3.21.6), anchor each section of each unit independently, as sectionalized for rail transport (see 3.21.3), to a flat car using only the tiedown attachments. Use no blocking.

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Hump the flatcar at a speed of 7 miles per hour against five box cars with their brakes set by air. Repeat the test with the opposite end of section in the direction of the five cars. Any permanent deformation or breakage of the tiedown attachments or parts of the section shall constitute failure of this test.

4.5.3.9 Production test. Each conveyor shall be run for 1 hour without load and towed for a distance of not less than 5 miles with lights on. Evidence of malfunction, permanent deformation, operational difficulties, or breakage shall consitute failure of this test.

4.5.3.10 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load, MIL-STD-1474, paragraph 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position, additionally, the noise level at the typical operating position shall be provided as db(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.6 Inspection comparison. The Government may select conveyors at any time during the contract production period and subject these conveyors to the examination specified in 4.5.2 and the test specified in 4.5.3 through 4.5.3.10 to determine that the quality of the selected conveyors is equal to the quality standards established during first article testing. The inspection will be performed by the Government at a site selected by the Government. Conveyors will be selected at random from those which have been accepted by the Government and will not include the previously inspected first article conveyor(s).

4.6.1 Inspection failure. Failure of any conveyor when inspected as specified in 4.6 may be considered cause for refusal by the Government to continue acceptance of conveyors until objective evidence furnished by the contractor reveals that corrective action has been taken to eliminate the condition which caused the rejection. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on conveyors yet to be produced by the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all conveyors accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer. Where conveyors previously accepted by the Government are determined to be deficient, the contractor will negotiate with the Government to determine the method of correction.

4.7 Inspection of packaging.

4.7.1 First article packaging inspection.

4.7.1.1 Examination. The first article pack shall be examined for the defects listed in 4.7.2.3. Presence of one or more defects shall be cause for rejecting the first article pack.

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4.7.1.2 Test. The first article pack of crated conveyors for level A (see 5.3.1 and 6.2), shall be subjected to the rail-impact test in accordance with MIL-STD-1186, appendix A. Criteria for failing the test shall be as specified therein.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Unit of product. For the purpose of inspection, a complete conveyor prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2, shall be examined for the following defects. AQL shall be 2.5 percent defective.

No.	Defect	A	B	Comm
130.	Sectionalization and disassembly not as specified.	5.2.1.2	5.2.2	5.2.3
131.	Matchmarking of removed parts and mating parts remaining on the conveyor not as specified.	5.2.1.2	5.2.2	
132.	Large components not stenciled for matchmaking as specified.	5.2.1.2	5.2.2	
133.	Small parts not tagged for matchmaking as specified.	5.2.1.2	5.2.2	
134.	Cloth tags not marked with waterproof ink as specified.	5.2.1.2	5.2.2	
135.	Unfinished surfaces not coated and wrapped as specified.	5.2.1.4.1	5.2.2	
136.	Machined surfaces not coated as specified.	5.2.1.4.2	5.2.2	
137.	Engine not preserved as specified.	5.2.1.5	5.2.2	
138.	Conveyor belt fasteners not coated, belt not removed, rolled and secured as specified.	5.2.1.6	5.2.2	
139.	Drive belts and pulleys not preserved as specified.	5.2.1.7	5.2.2	
140.	Drive belts not removed or released from tension as specified.	5.2.1.7	5.2.2	
141.	Removed belts and pulley grooves not preserved as specified.	5.2.1.7	5.2.2	
142.	Instruction tag requirements for drive belts not as specified.	5.2.1.7	5.2.2	
143.	Drive shafts and universal joints not preserved as specified.	5.2.1.8	5.2.2	
144.	Exposed gears not preserved as specified.	5.2.1.9.1	5.2.2	

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145.	Enclosed gears not preserved as specified.	5.2.1.9.2	5.2.2	
146.	Exposed drive chains not preserved as specified.	5.2.1.10.1		
147.	Enclosed drive chains not preserved as specified.	5.2.1.10.2	5.2.2	
148.	Hydraulic elevating mechanism not preserved as specified.	5.2.1.11.1	5.2.2	
149.	Mechanical elevating mechanism parts not preserved as specified.	5.2.1.11.2	5.2.2	
150.	Wheels and tires not preserved as specified.	5.2.1.12	5.2.2	
151.	Repair parts not preserved as specified.	5.2.1.13	5.2.2	
152.	Maintenance tools not preserved as specified.	5.2.1.14	5.2.2	
153.	Technical publications not preserved as specified.	5.2.1.15	5.2.2	
154.	Toolbox lid not closed and secured as specified.	5.2.1.16	5.2.2	
155.	Brake system not preserved as specified.	5.2.1.17	5.2.2	
156.	Sockets and coupling not protected for crafted shipment as required.	5.2.1.18	5.2.2	
157.	Lights/lamps and reflectors not preserved as specified for crated shipment.	5.2.1.19	5.2.2	
158.	Clutch and clutch control not preserved as specified.	5.2.1.20	5.2.2	
159.	Control levers not preserved as specified.	5.2.1.21	5.2.2	
160.	Consolidation not as specified.	5.2.1.23	5.2.2	
161.	Commercial preservation not as specified.			5.2.3
162.	Conveyor not prepared for unboxed (mobile) shipment as specified.	5.3.1.1		5.3.2
163.	Conveyor not packed as specified for crated shipments.	5.3.1.2		5.3.2
164.	Marking not as specified.	5.4.1		5.4.2

5. PACKAGING

5.1 First article packaging. Unless otherwise specified (see 6.2), the contractor shall furnish a first article pack for examination and testing within the time frame specified (see 6.2), to prove prior to the start of production packaging, that the applied preservation, packing and marking comply with the requirements of this specification. Examination and testing shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.6). The first article pack may utilize either the first article model or a production model. When the first article model is utilized,

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and the Government requests a comparison between the first article model and a production model, any preservation and packing shall be removed by the contractor at no expense to the Government.

5.2 Preservation. Preservation shall be level A, level B or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Sectionalization and disassembly. The conveyor shall be sectionalized to meet the applicable requirements of 3.21.3 or 3.21.4. In addition, components of the conveyor and sections vulnerable to damage, pilferage, or loss, and protruding parts that increase cubage and that can be removed and reassembled without special tools shall be removed. Removed bolts, nuts, pins, and washers shall be replaced in one of the mating parts and secured to prevent their loss. Removed keys shall be secured in the keyway of the pulley or shaft for which intended.

5.2.1.2 Matchmarking. To facilitate reassembly, components and parts removed from the conveyor and the sections shall be matchmarked. Large parts shall be matchmarked by stenciled letters or numerals with lusterless white enamel, overcoated with varnish conforming to TT-V-121 or with paint conforming to MIL-P-13983. Small parts shall have matchmarking information on cloth tags conforming to UU-T-81, type A, or on metal tags attached to mating parts. The tags shall be securely attached to the parts in such manner as to prevent damage to the preservation or to the parts. The cloth tags shall be marked with waterproof ink.

5.2.1.3 Preservatives. The provisions of MIL-P-116 shall apply when determining the applicability and when selecting preservatives. Preservatives specified herein by "P" numbers shall conform to the applicable specifications listed in MIL-P-116 and shall be applied in accordance with MIL-P-116.

5.2.1.4 Unprotected surfaces.

5.2.1.4.1 Unfinished surfaces. Unpainted exterior metal surfaces of items, components, parts, and assemblies, such as bolts, nuts, washers, rolls, sprockets, shifting takeup adjusting rods, spring, chock blocks, safety chains, coupling pins, towing hooks, and any other exterior metal surfaces not specified herein, requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.4.2 Machined surfaces. Exposed exterior machined surfaces shall be coated with type P-6 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2 and secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.5 Engine. The engine and engine accessories shall be preserved in accordance with MIL-E-10062, level A as specified for type I or II classification as applicable, method I, except that cooling systems for liquid cooled engines shall be preserved with antifreeze conforming to MIL-A-46153.

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5.2.1.6 Conveyor belt. The conveyor belt shall be removed. Belt fasteners shall be coated with varnish conforming to TT-V-121. The belt shall be compactly rolled and the roll shall be secured with strapping conforming to QQ-S-781, type I or IV, heavy duty, class B, size as applicable. The strapping shall be placed circumferentially on the roll and the straps tensioned so as not to damage the belt.

5.2.1.7 Drive belts and pulleys. Belts shall be removed or released from tension. Unpainted pulley grooves shall be coated with primer conforming to TT-P-664. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3. A tag conforming to UU-T-81, type A, shall be attached in a conspicuous location indicating:

"BELTS HAVE BEEN REMOVED OR RELEASED FROM TENSION. INSTALL OR TENSION AS APPLICABLE, PRIOR TO OPERATION."

The information on the tag shall be applied with waterproof ink.

5.2.1.8 Drive shafts and universal joints. Unpainted metal surfaces of drive shafts and universal joints shall be coated with type P-1 preservative. Dust boots, if furnished, shall be reinstalled.

5.2.1.9 Gears.

5.2.1.9.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664.

5.2.1.9.2 Enclosed gears. Gears operating on lubricating oil (SAE 10, 30 or 50) shall have the housing filled to the operating level with type P-10 preservative, type I, grade 10, 30, or 50 as applicable and tagged to indicate: "The preservative oil is adequate for operation. Do not drain until the first required lubrication change." Gears operating on gear lubricant (SAE 80 or 90) shall be filled to the operating level with lubricant conforming to MIL-L-2105, grade as applicable, and tagged to indicate: "The lubricant is good for operation. Do not drain until the first required lubrication change." Gears not operating on lubricating oil (SAE 10, 30, or 50) or lubricant (SAE 80 or 90) shall be filled to the operating level with the approved lubricant required for operation. The gear housings shall be tagged to indicate: "The housing is filled to the operating level with the applicable type lubricant. Do not drain prior to the first required lubrication change." The tags shall conform to UU-T-81, type A. Information on the tags shall be applied with waterproof ink. The tags shall be attached in a conspicuous location.

5.2.1.10 Drive chains.

5.2.1.10.1 Exposed chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. The excess preservative shall be drained and the chain recoated with type P-1 preservative. Removed

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chains shall be coiled and the coil secured with soft annealed wire and wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured with tape conforming to PPP-T-60, type IV. Sprockets and other unprotected surfaces of the drives shall be coated with type P-1 preservative.

5.2.1.10.2 Enclosed chains. Enclosed chains and chain housing shall be preserved as specified in 5.2.1.9.2 for enclosed gears.

5.2.1.11 Elevating mechanism.

5.2.1.11.1 Hydraulic. The hydraulic fluid supply tank shall be filled to the operating level with the hydraulic fluid recommended for operation. The rams shall be fully retracted into the cylinders. Any exposed portion of the rams shall be coated with type P-6 prservative and wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, extending the wrap approximately 2 inches on the ram cylinder. The wrap shall be secured in place with tape conforming to PPP-T-60, type IV. The control valve shall be closed. Any other metal surface of components and parts of the system requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.11.2 Mechanical. The wire rope shall be coated with type P-1 preservative. Other components of the mechanical elevating mechanism shall be preserved as specified herein for similar components.

5.2.1.12 Wheels and tires. When wheels are removed, the tires shall be inflated to two-thirds of the specified operating pressure. The wheel nuts shall be replaced on the studs and the exposed threaded surfaces shall be coated with type P-1 preservative. When wheels are not removed, the tires shall be inflated to 10 pounds above the specified operating pressure.

5.2.1.13 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve, repair parts, and the repair parts placed in the toolbox if space permits.

5.2.1.14 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40, level A and placed in the toolbox if space permits.

5.2.1.15 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3 and placed in the toolbox.

5.2.1.16 Toolbox. The toolbox lid shall be closed and secured.

5.2.1.17 Brake system (crated shipments). In addition to the preservation specified in 3.20.3, the air-actuated brake system shall be drained. Interior surfaces of the air-supply tanks shall be fogged with type P-10 preservative, type I, grade 30. Excess preservative oil shall be drained. Threaded openings and the threads of drain plugs shall be coated with the preservative oil specified herein. The drain plugs shall be reinstalled. Air line filters shall be

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drained and closed. The exhaust ports of relay emergency, quick-release, and relay valves not equipped with exhaust check valves shall be closed by inserting caps or plugs as applicable conforming to MIL-C-5501 or by sealing the ports with tape conforming to PPP-T-60, type IV. Tags conforming to UU-T-81, type A, shall be attached to the valves and service lines indicating: "The plugs, caps, or tape must be removed from the exhaust ports and valves before this roller is placed in operation." Waterproof ink shall be used for printing the information on the tags.

5.2.1.18 Sockets and couplings (crated shipments). When the conveyor is to be packed in a crate (see 6.2), the openings in sockets and couplings shall be sealed by the means provided or with tape conforming to PPP-T-60, type IV.

5.2.1.19 Lights/lamps and reflectors (crated shipments). When the conveyor is to be packed in a crate (see 6.2), removable light/lamp clusters shall be individually preserved in accordance with MIL-P-116, method III, using boxes conforming to PPP-B-636, W5c or W6c, style optional. Lights/lamps and reflectors not removed shall be covered with barrier material conforming to MIL-B-121, type II, grade A, class 2. The barrier material shall be secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.20 Clutch, and clutch control. With the cover plates removed and the clutch engaged, all accessible metal interior components of the clutch shall be sprayed with a thin film of primer conforming to TT-P-664 or primer conforming to MIL-P-46093. Clutch control mechanisms not enclosed shall be coated with type P-1 preservative. After being sprayed, the clutch control mechansim shall be secured in such position so as to completely disengage the disc facings and pressure plates. Cover plates shall be reinstalled.

5.2.1.21 Control levers. Unpainted metal surfaces of handles and levers shall be coated with type P-1 preservative. Composition and rubber handgrips shall be removed if practical and preserved in accordance with MIL-P-116, method IC-1 or IC-3 and placed in toolbox. When the handgrips are left in place, they shall be wrapped with barrier material conforming to MIL-B-121, type II, grade A, class 2 or black polyethylene conforming to L-P-378 (6 mil) secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.22 Other components. Other components not specifically mentioned herein requiring protection from corrosion of physical or mechanical damage shall be preserved as specified for components of similar design and construction.

5.2.1.23 Consolidation. Loose components requiring the protection of a container to prevent pilferage, damage or loss, that cannot be placed in the toolbox, secured to the conveyor or to the sections, shall be consolidated together in a minimum number of close-fitting boxes conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional. The number, size, weight and configuration of the boxes shall be determined by the available space and convenience for securing the boxes to the conveyor.

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5.2.2 Level B. Level B preservation shall be as specified in 5.2 for level A except that boxes specified in 5.2.1.19 and 5.2.1.23 shall be domestic type or ss.

5.2.3 Commercial. Commercial preservation shall be in accordance with ASTM D 3951 except that sectionalization and disassembly shall be as specified in 5.2.1.1.

5.3 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Unboxed (mobile) shipments. Unless otherwise specified (see 6.2), each complete conveyor, preserved as specified in 5.2, shall be prepared for unboxed (mobile) shipment. Consolidation containers, and components not suitable for placement in the consolidation containers or toolbox, shall be secured to the basic unit so as not to interfere with lifting the conveyor or with its mobility. The securing shall be accomplished by the application of (a) steel strapping conforming to QQ-S-781, type I, or IV heavy duty, finish b, (b) bolts or (c) a combination of steel strapping and bolts. The size, placement and quantity of such securing devices shall be sufficient to assure positive containment of the components in the desired configuration. Blocking, bracing and cushioning shall be applied, as required, to accomplish that task.

5.3.1.2 Crated shipments. When specified, each complete conveyor, preserved as specified in 5.2, shall be packed in a crate conforming to MIL-C-3774, type style A. Blocking, bracing, anchoring and cushioning shall be as specified the appendix thereto and in accordance with MIL-STD-1186.

5.3.2 Commercial. Unless otherwise specified (see 6.2), each complete conveyor, preserved as specified in 5.2, shall be prepared for unboxed (mobile) shipment in a manner to assure carrier acceptance and safe delivery to destination at lowest rates in complicance with the rules and regulations for the mode of transportation selected. When conveyors are specified to be crated (see 6.2), each complete conveyor, preserved as specified in 5.2, shall be packed in accordance with ASTM D 3951.

5.4 Marking.

5.4.1 Levels A and B. In addition to any special or identification markings required by the contract or purchase order, each conveyor preserved and packed in accordance with the applicable provisions of levels A and B, shall be marked in accordance with MIL-STD-129.

5.4.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. Additionally, each shipping container and, as applicable, each unboxed (mobile) conveyor shall be marked with the gross weight and cube.

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6. NOTES

6.1 Intended use. The conveyor is intended for use in handling sand, gravel, crushed rock, coal, hot-mix bituminous material, and similar bulk material.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of the first article model (see 3.2).
c. When a discharge hood with fixed spout is required on the head-end section (see 3.12).
d. When a swivel spout is required (see 3.12).
e. Time frame for submission of certification.
f. Technical publications required (see 3.32).
g. Applicable list of repair parts and maintenance tools (see 3.33).
h. When the 24-volt electrical system for military truck compatability is not required (see 3.20.6.1).
i. When a belt for handling hot asphaltic mixes is required (see 3.13).
j. When a belt housing is required (see 3.15).
k. Time interval for submission of electromagnetic interference report (see 4.5.3.4).
l. When a first article pack is not required (see 5.1).
m. Time frame for submission of first article pack when required (see 5.1).
n. Degree of preservation and degree of packing required (see 5.2 and 5.3 respectively).
o. When conveyors are to be crated (see 5.3.1.2 and 5.3.2).
p. Any special marking requirements (see 5.4).
q. Provisioning requirements (see 6.8).
r. Color required when other than as specified (see 3.29).
s. Time frame when camouflage pattern data (see 3.37) is required.

6.2.1 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1644) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data requirements	Applicable DID No.
a. 3.37	Camouflage pattern photos.	UDI-V-24003B/M (see figure 1)
b. 3.37	Camouflage pattern illustrations.	DI-M-30420/M (see figure 2)

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DATA ITEM DESCRIPTION	2 IDENTIFICATION NO(S).	
	AGENCY	NUMBER
1. TITLE PHOTOGRAPHIC PRINTS AND SLIDES	NAVY-YD	UDI-V-24003B/M

3. DESCRIPTION/PURPOSE

This data consists of photographic prints and/or slides which provide a pictorial medium for exchange of technical information between developing Agency and users and becomes a permanent historical and legal record by recording configuration during the course of program development.

4. APPROVAL DATE: 13 June 1972

5. OFFICE OF PRIMARY RESPONSIBILITY: NAVFAC 0434

6. DDC REQUIRED

8. APPROVAL LIMITATION

7. APPLICATION/INTERRELATIONSHIP

7.1 Deliverable data covered by this DID and specified in block 16 of the CDRL, is required as a visual record of equipment procured under contract.

9. REFERENCES (Mandatory as cited in block 10)

MCSL NUMBER(S)

PREPARATION INSTRUCTIONS

10.1 These photographs shall be standard 8x10 inch prints and furnished in 2x2 inch cardboard frames; and unless otherwise specified on the DD Form 1423. Views shall include, but not be limited to, a front, side, and rear view. Background of photographs shall be void of unrelated objects.

10.2 Interior views shall be in sufficient quantity to show arrangement and layout of installed equipment.

10.3 Photographic prints shall show excellent detail and shall be color suitable for reproduction.

DD FORM 1664

FIGURE 1. Camouflage pattern photos.

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DATA ITEM DESCRIPTION	2 IDENTIFICATION NO(S).	
	AGENCY	NUMBER
1. TITLE Technical Illustrations and Graphs	USAF	DI-M-30420/M
3. DESCRIPTION/PURPOSE To provide wave plotting, drafting, computations, data reduction and/or mathematical analysis in support of research projects.	4. APPROVAL DATE 29 Oct 1976	
	5. OFFICE OF PRIMARY RESPONSIBILITY AFSC	
	6. DDC REQUIRED	
	8. APPROVAL LIMITATION	
7. APPLICATION/INTERRELATIONSHIP Mathematical Services Projects	9. REFERENCES (Mandatory as cited in block 10)	
Replaces (U)M-177/ESD	MCSL NUMBER(S)	

10. PREPARATION INSTRUCTIONS

Data will encompass plotting, drafting and artistic creative illustrations of apparatus and devices. The work may include three-dimensional pictorial representations. Completed reproducible art work must be of sufficiently high quality to be used for publication in technical journals and for visual aids at conferences, reviews and briefings.

DD FORM 1664

FIGURE 2. Camouflage pattern illustrations.

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6.3 First article. When a first article inspection is required, the items should be a first item produced. The first article should consist of 1 unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article test results and disposition of the documents' first article.

6.4 Lubricants. Selection of lubricants shall be in accordance with section 5 of MIL-STD-838. Unless otherwise authorized by the US Army, Belvoir R&D Center, lubricants for ground equipment systems shall be restricted to those limited in MIL-HDBK-113.

6.5 Government-loaned equipment. The contracting officer should arrange to loan the dump truck listed in 3.34, and to furnish the property listed in 3.37.

6.6 First article pack. Any changes or deviations of production packs from the approved first article pack during production will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack and mark the units in accordance with this specification.

6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each conveyor.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of conveyors. A suggested paragraph is as follows:

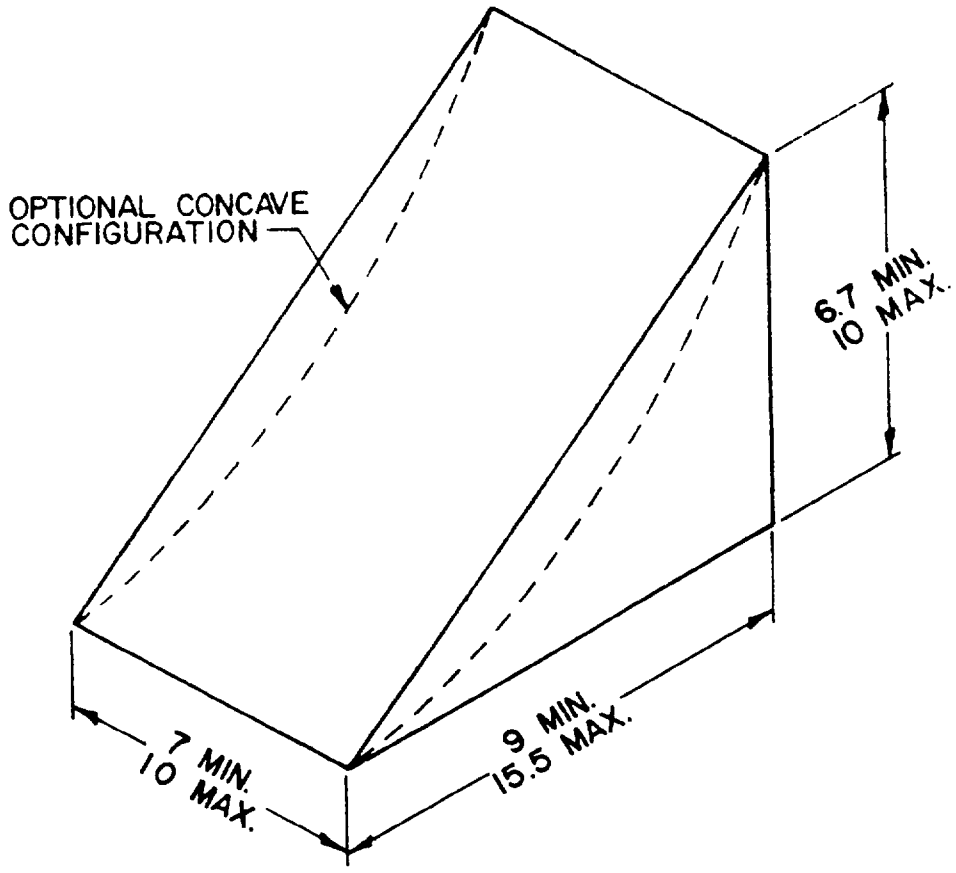
"Shipment of conveyors shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 3910-A072

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CHOCK MATERIAL NOTE

. ALUMINIUM
2. MAGNESIUM
3. SEMI-RIGID ELASTOMER
4. MAXIMUM CHOCK WEIGHT- 11.5 LB.

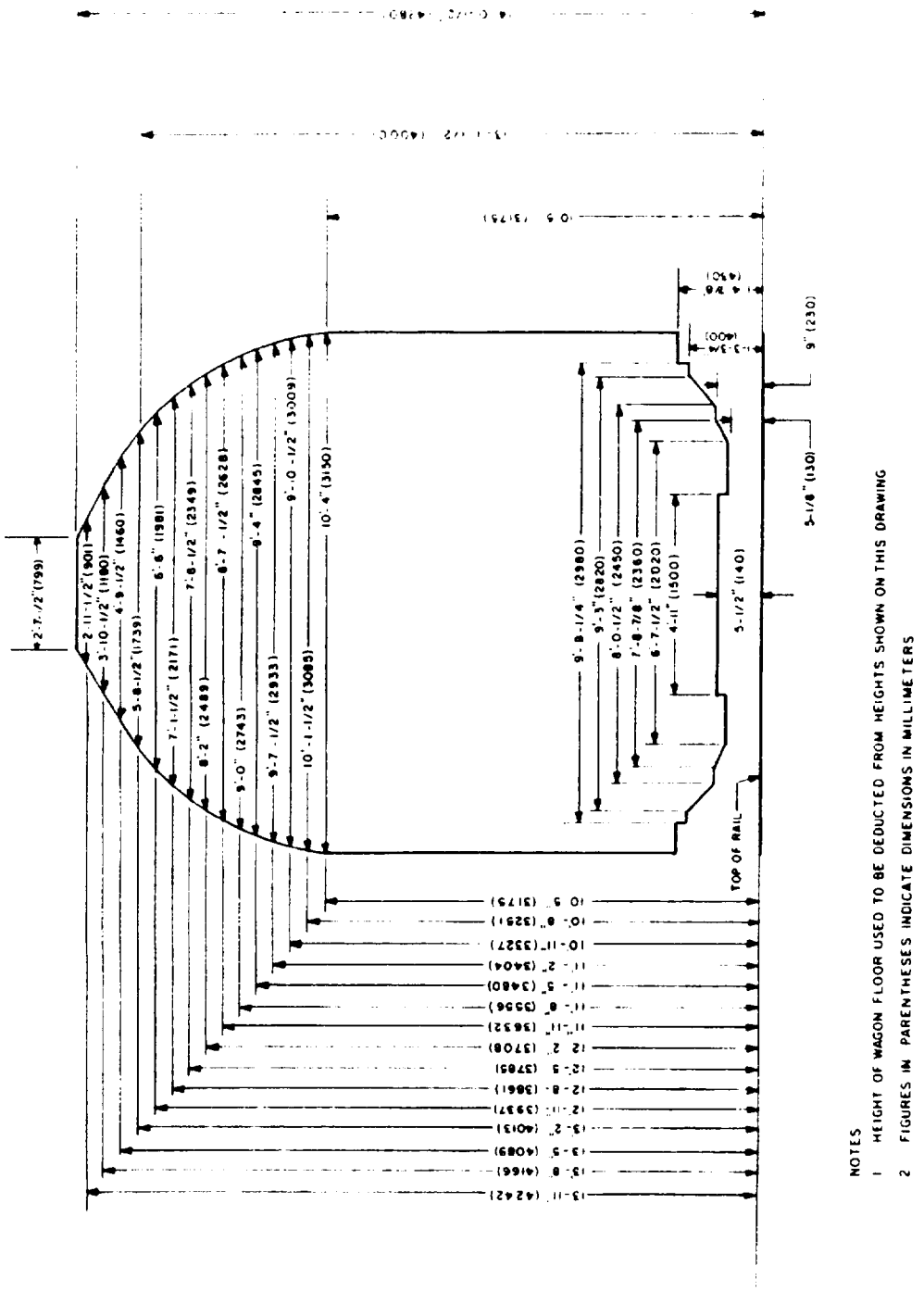
CAPTIVE FASTENER ASSEMBLY DETAILS

1. SNAP HOOK FOR CRADLE OR EYE.
2. FASTENER LINE TENSILE STRENGTH MINIMUM OF 1,400 LB.
3. PLATE FERROUS COMPONENTS TO RESIST CORROSION.
4. LENGTH OF CAPTIVE FASTENER ASSEMBLY SHALL REACH BETWEEN CRADLE OR HANGER AND EMPLACED CHOCK BLOCK AT EXTREME LOCATIONS AHEAD OF AND BEHIND ROAD WHEELS.

FIGURE 3. Chock block assembly.

X 603 A

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NOTES

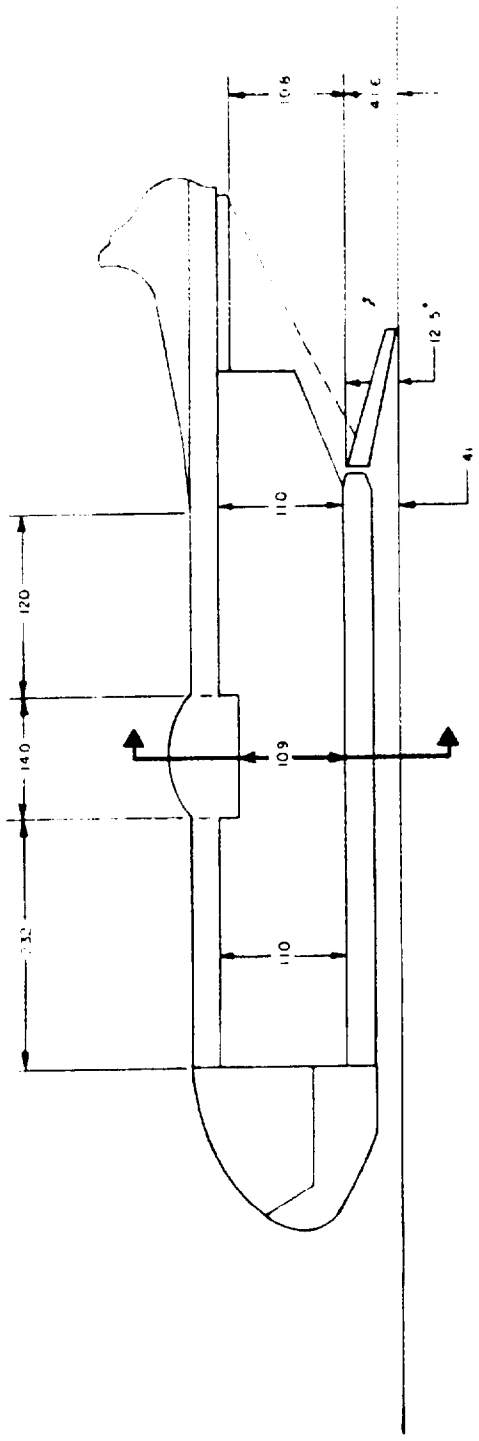
1 HEIGHT OF WAGON FLOOR USED TO BE DEDUCTED FROM HEIGHTS SHOWN ON THIS DRAWING

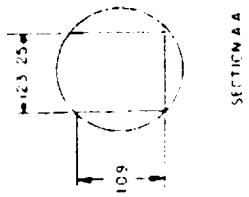
2 FIGURES IN PARENTHESES INDICATE DIMENSIONS IN MILLIMETERS

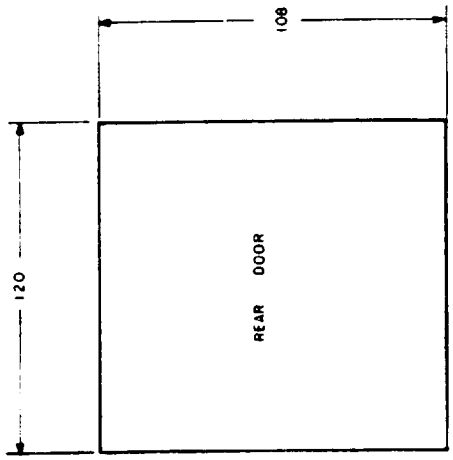
FIGURE 4. International loading gauge.

X-3605A

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NOTE UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES

FIGURE 5. Cargo compartment profile (C-130).

X-195B

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions — Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-3972H (ME)	CONVEYOR, BELT; PORTABLE

a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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